

Credit Name: File Structures & Exception Handling
Assignment Name: Countvowels.java

The following questions can help you in thinking critically about your problem-solving processes:

Understanding the Problem:

How did you approach understanding the challenge?

I approached understanding the challenge by first figuring out what each thing meant in the template that was provided so I can flawlessly write the code. When I didn't understand I would ask another friend from computer science 30 or chatgpt. Asking them allowed me to get through the understanding process faster because if I tried learning it alone I would have used up a lot of time.

```
/* count the vowels in the file */  
try {  
    //new File object required  
    //new File reader object required  
    //new BufferedReader object required  
  
    //read the lines from the file  
    //as lines are read convert them to lower case  
    //iterate through the lower case text  
        //using the String class  
        //check if each character traverse through is a vowel  
        //update total vowels
```

I was using ChatGpt to understand this part of the code because I didn't have a visual idea on how to make it step by step.

Were there any parts of the problem you found confusing at first? If so, how did you resolve that confusion?

Something that Confused me at first was a line that said `/* prompt the user for the name of the file */`. This confused me because I thought I had to make the user generate a new file but that wouldn't make sense because there are no vowels imputed in the vowel. I resolved the confusion by asking Mr.Aballa and he said the user will have to put it in the file that was created by me.

```
/* prompt the user for the name of the file */  
//????
```

Planning the Solution

Did you create a plan or break the problem into smaller steps before coding?

What I did was break each part into smaller steps and use the template and write each line of code while testing it when I make another function. Breaking it into smaller steps allowed me to be organized and allowed me to follow the template without being stuck.

```
public class CountVowels {  
  
    public static void main(String[] args) {  
  
        File dataFile; // declares a file variable (not used here)  
        File textFile; // stores the file the user wants to open  
        FileReader in; // reads characters from the file  
        BufferedReader readFile; // reads full lines from the file  
        String fileName; // stores the filename entered by the user  
        String lineInFile, lowercaseText, letter; // stores each line, lowercase line,  
        int vowelSum = 0; // counts total vowels  
        Scanner input = new Scanner(System.in); // creates scanner for user input  
  
        System.out.println("Enter the file name: "); // asks user for filename  
        fileName = input.nextLine(); // stores the typed filename
```

I did this part first because I understood that without asking for the user to enter a file the whole code wouldn't work because the vowels are inputted inside the file and the program is supposed to read the vowels.

```
        textFile = new File(fileName); // creates a file object using the filename  
        if (textFile.exists()) { // checks if the file exists  
            System.out.println("File exists"); // tells user file was found  
            System.out.println("Gathering vowels"); // lets user know counting started  
        } else { // runs if file does not exist  
            System.out.println("File does not exist"); // tells user file was not found  
        }  
    }  
}
```

I had to do this part in another step because this is where the program finds the file and checks if the file exists or not.

```

in = new FileReader(textFile); // opens the file for reading

readFile = new BufferedReader(in); // connects buffered reader to file reader

while ((lineInFile = readFile.readLine()) != null) { // keeps reading until end of file
    lowercaseText = lineInFile.toLowerCase(); // changes line to lowercase

    for (int i = 0; i < lowercaseText.length(); i++) { // loops through each character
        letter = lowercaseText.substring(i, i + 1); // gets one letter at a time

        if (letter.equals("a") || letter.equals("e") || // checks if letter is a vowel
            letter.equals("i") || letter.equals("o") ||
            letter.equals("u")) {
            vowelSum++; // adds 1 if vowel is found
        }
    }
}

System.out.println("The file has " + vowelSum + " vowels."); // shows final total

```

For this part I used a little bit of chatgpt and I came after the spring break making my mind foggy and forget what I had to do. At the end I was able to complete it and understand it allowing me to do it again if it was given

How did you decide on the tools, data structures, or algorithms to use?

I chose the algorithm as a linear scan because the goal of the program is to count every vowel in the file by checking each character exactly once. After reading each line from the file, the program uses a for loop to move through the text from the first character to the last, which makes the logic simple and efficient. For every position in the String, it checks whether the character is one of the vowels (a, e, i, o, u) and increases the counter if it is. This method works well because it guarantees that no characters are skipped, it is easy to understand.

```

if (letter.equals("a") || letter.equals("e") || //
    letter.equals("i") || letter.equals("o") ||
    letter.equals("u")) {
    vowelSum++; // adds 1 if vowel is found
}

```

Implementation

Did you write the code in small pieces or attempt the entire solution at once?

I wrote the code in small pieces because the template allowed me to have an idea before I made the code allowing me to make it flawlessly without being interrupted with an issue that will slow me down in making the program.

How did you test your solution along the way to make sure it was working?

I tested my code along the way to make sure each step was working because it allowed me to catch any error before something I made an extremely large error which makes me restart the whole code. The biggest issue in restarting is time and because for the last credit I took a lot of time and I was late so I decided to make sure there was no error at all when I fully completed it.

```
Enter the file name:
```

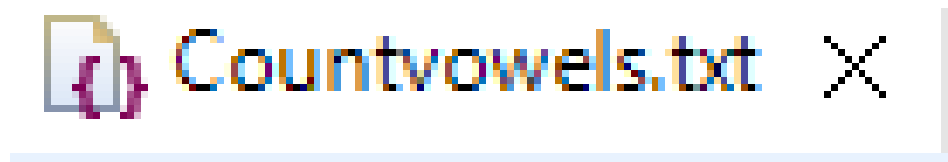
```
|
```

I had to make sure this part was able to work or the whole code didn't work and allowing me to do it step by step made me make sure that there was no error involved in making the code.

Overcoming Challenges

What part of the problem was the most difficult for you?

The problem that was most difficult for me was understanding how the program reads the file. By looking at the skill builders and past lessons, I was able to understand that the program first looks for the file using the file path, then opens it and reads the contents line by line. I also learned that it only works if the correct file name is entered, and for text files that usually means including ".txt" at the end of the file name so the program can find the right file.



This is my file that the program reads and extracts the information

How did you handle moments when you felt stuck or unsure of what to do next?

When I was stuck I would re-read the template and see what I would have to do and then I would ask a friend and finally chatgpt. Getting insight from everything allows me to understand the problems which allowed me to fix it and keep going.

Learning

Was there anything you learned that you think will help you with future challenges?

Something I learned that will help me with future challenges is the theory I learned allowing me to apply file structures when I think it is the appropriate time. Many times I was stuck because of the theory involved because I couldn't wrap my head around some of it and I had to learn it and understand it through many ways.